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# CAN Protocol

CAN Protocol Instructions for use：

The serial port sending command must be completed within 10S, otherwise it will be

automatically locked. In order to avoid automatic locking, the following steps can be performed first.

1. Enter the unlock command
2. Enter the command that needs to modify or read the data
3. Save the command

Register table

| AD DR (He x) | AD DR (De c) | REGISTE R NAME | FUNCTION | SER IAL I/F | Bit 15 |
|---|---|---|---|---|---|
| 00 | 00 | SAVE | save/reboo t/ factory reset | R/W | SAVE[15:0] |
| 01 | 01 | CALSW | Calibration mode | R/W | |
| 02 | 02 | RSW | output content | R/W | |
| 03 | 03 | RRATE | output rate | R/W | |
| 04 | 04 | BAUD | Serial port baud rate | R/W | |
| 05 | 05 | AXOFFSE T | Acceleratio n X Zero Bias | R/W | AXOFFSET [15:0] |

| | | | | | |
|---|---|---|---|---|---|
| 06 | 06 | AYOFFSET | Acceleration Y Zero Bias | R/W | AYOFFSET[15:0] |
| 07 | 07 | AZOFFSET | Acceleration Z Zero Bias | R/W | AZOFFSET[15:0] |
| 08 | 08 | GXOFFSET | Angular velocity X zero bias | R/W | GXOFFSET[15:0] |
| 09 | 09 | GYOFFSET | Angular velocity Y zero bias | R/W | GYOFFSET[15:0] |
| 0A | 10 | GZOFFSET | Angular velocity Z zero bias | R/W | GZOFFSET[15:0] |
| 0B | 11 | HXOFFSET | Magnetic Field X Zero Bias | R/W | HXOFFSET[15:0] |
| 0C | 12 | HYOFFSET | Magnetic Field Y Zero Bias | R/W | HYOFFSET[15:0] |
| 0D | 13 | HZOFFSET | Magnetic Field Z Zero Bias | R/W | HZOFFSET[15:0] |
| 1A | 26 | IICADDR | Device address | R/W | |
| 1B | 27 | LEDOFF | Turn off LED light | R/W | |
| 1C | 28 | MAGRANGX | Magnetic Field X Calibration Range | R/W | MAGRANGX[15:0] |
| 1D | 29 | MAGRAN | Magnetic | R/W | MAGRANGY[15:0] |

| | | | | | |
|---|---|---|---|---|---|
| | | GY | Field Y Calibration Range | | |
| 1E | 30 | MAGRANGZ | Magnetic Field Z Calibration Range | R/W | MAGRANGZ[15:0] |
| 1F | 31 | BANDWIDTH | bandwidth | R/W | |
| 20 | 32 | GYRORANGE | Gyroscope range | R/W | |
| 21 | 33 | ACCRANGE | Acceleration range | R/W | |
| 22 | 34 | SLEEP | sleep | R/W | |
| 23 | 35 | ORIENT | Installation direction | R/W | |
| 24 | 36 | AXIS6 | Algorithms | R/W | |
| 25 | 37 | FILTK | Dynamic filtering | R/W | FILTK[15:0] |
| 27 | 39 | READADDR | Read registers | R/W | |
| 2A | 42 | ACCFILT | Acceleration filtering | R/W | ACCFILT[15:0] |
| 2D | 45 | POWONSEND | command start | R/W | |
| 2E | 46 | VERSION | version number | R | VERSION[15:0] |
| 30 | 48 | YYMM | year/mouth | R/W | MOUTH[15:8] |

| | | | | | |
|---|---|---|---|---|---|
| 31 | 49 | DDHH | day/hour | R/W | HOUR[15:8 ] |
| 32 | 50 | MMSS | minute/sec o nds | R/W | SECONDS[ 15:8] |
| 33 | 51 | MS | millisecond | R/W | MS[15:0] |
| 34 | 52 | AX | Acceleratio n X | R | AX[15:0] |
| 35 | 53 | AY | Acceleratio n Y | R | AY[15:0] |
| 36 | 54 | AZ | Acceleratio n Z | R | AZ[15:0] |
| 37 | 55 | GX | Angular velocity X | R | GX[15:0] |
| 38 | 56 | GY | Angular velocity Y | R | GY[15:0] |
| 39 | 57 | GZ | Angular velocity Z | R | GZ[15:0] |
| 3A | 58 | HX | Magnetic Field X | R | HX[15:0] |
| 3B | 59 | HY | Magnetic field Y | R | HY[15:0] |
| 3C | 60 | HZ | Magnetic field Z | R | HZ[15:0] |
| 3D | 61 | Roll | Roll angle | R | Roll[15:0] |
| 3E | 62 | Pitch | Pitch angle | R | Pitch[15:0] |
| 3F | 63 | Yaw | Heading | R | Yaw[15:0] |
| 40 | 64 | TEMP | temperatur e | R | TEMP[15:0 ] |

| | | | | | |
|---|---|---|---|---|---|
| 45 | 69 | PressureL | Air pressure low 16 bits | R | PressureL[15:0] |
| 46 | 70 | PressureH | Air pressure high 16 bits | R | PressureH[15:0] |
| 47 | 71 | HeightL | Height lower 16 bits | R | HeightL[15:0] |
| 48 | 72 | HeightH | High low high 16 bits | R | HeightH[15:0] |
| 51 | 81 | q0 | Quaternion 0 | R | q0[15:0] |
| 52 | 82 | q1 | Quaternion 1 | R | q1[15:0] |
| 53 | 83 | q2 | Quaternion 2 | R | q2[15:0] |
| 54 | 84 | q3 | Quaternion 3 | R | q3[15:0] |
| 61 | 97 | GYROCALITHR | Gyro Still Threshold | R/W | GYROCALITHR[15:0] |
| 62 | 98 | ALARMLEVEL | Angle alarm level | R/W | |
| 63 | 99 | GYROCALTIME | Gyro auto calibration time | R/W | GYROCALTIME[15:0] |
| 68 | 104 | TRIGTIM E | Alarm continuous trigger time | R/W | TRIGTIME[15:0] |
| 69 | 105 | KEY | Unlock | R/W | KEY[15:0] |
| 6A | 106 | WERROR | Gyroscope | R | WERROR[15:0] |

| | | | | | |
|---|---|---|---|---|---|
| | | | change value | | |
| 6E | 110 | WZTIME | Angular velocity continuous rest time | R/W | WZTIME[15:0] |
| 6F | 111 | WZSTATI C | Angular velocity integral threshold | R/W | WZSTATIC [15:0] |
| 79 | 121 | XREFROL L | Roll angle zero reference value | R | XREFROLL [15:0] |
| 7A | 122 | YREFPIT CH | Pitch angle zero reference value | R | YREFPITCH[15:0] |
| 7F | 127 | NUMBERI D1 | Device No.1-2 | R | ID2[15:8] |
| 80 | 128 | NUMBERI D2 | Device No. 3-4 | R | ID4[15:8] |
| 81 | 129 | NUMBERI D3 | Device No. 5-6 | R | ID6[15:8] |
| 82 | 130 | NUMBERI D4 | Device No. 7-8 | R | ID8[15:8] |
| 83 | 131 | NUMBERI D5 | Device No. 9-10 | R | ID10[15:8] |
| 84 | 132 | NUMBERI D6 | Device No. 11-12 | R | ID12[15:8] |

## Protol format

### Active Output Format

- The data is sent in hexadecimal not ASCII。
- Each data is transmitted in sequence by low byte and high byte, and the two are combined into a signed short type of data. For example, for data DATA1, DATA1L is the low byte and DATA1H is the high byte. The conversion method is as follows: Suppose DATA1 is the actual data, DATA1H is its high-byte part, DATA1L is its low-byte part,

Then: DATA1=(short)((short)DATA1H<<8|DATA1L). It must be noted here that DATA1H needs to be coerced into a signed short type of data before shifting, and the data type of DATA1 is also a signed short type, so that negative numbers can be represented.

| Protocol headers | Data Content | Data lower 8 bits | Data high 8 bits | Data lower 8 bits | Data high 8 bits |
|---|---|---|---|---|---|
| 0x55 | TYPE 【1】 | DATA1L[7:0] | DATA1H[15:8] | DATA2L[7:0] | DATA2H[15:8] |

【1】TYPE(Data content):

| TYPE | Remark |
|---|---|
| 0x50 | Time |
| 0x51 | Acceleration |
| 0x52 | Angular velocity |
| 0x53 | Angle |
| 0x54 | Magnetic field |
| 0x5F | Read |

| 0x55 | 0x50 | YY | MM | DD | HH |
|---|---|---|---|---|---|
| Name | Description | Remark | | | |
| YY | Year | | | | |
| MM | Mouth | | | | |
| DD | Day | | | | |
| HH | Hour | | | | |
| MN | Munite | | | | |
| SS | Seconds | | | | |

| 0x55 | 0x51 | AxL | AxH | AyL | AyH |
|---|---|---|---|---|---|
| Name | Descrption | Remark | | | |
| AxL | Acceleration X low 8 bits | Acceleration X= ((AxH<<8)\|AxL)/32768*16g<br>(g is the acceleration of gravity, preferably 9.8m/s2) | | | |
| AxH | Acceleration X high 8 bits | | | | |
| AyL | Acceleration Y low 8 bits | Acceleration Y= ((AyH<<8)\|AyL)/32768*16g<br>(g is the acceleration of gravity, preferably 9.8m/s2) | | | |
| AyH | Acceleration Y high 8 bits | | | | |
| AzL | Acceleration Z low 8 bits | Acceleration Z= ((AzH<<8)\|AzL)/32768*16g | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | (g is the acceleration of gravity, preferably 9.8m/s2) | | | |
| AzH | Acceleration Z high 8 bits | | | | |

| 0x55 | 0x52 | WxL | WxH | WyL | WyH |
|---|---|---|---|---|---|
| Name | Description | Remark | | | |
| WxL | Angular velocity X low 8 bits | Angular velocity X=((WxH<<8)\|WxL)/32768*2000°/s | | | |
| WxH | Angular velocity X high 8 bits | | | | |
| WyL | Angular velocity Y low 8 bits | Angular velocity Y=((WyH<<8)\|WyL)/32768*2000°/s | | | |
| WyH | Angular velocity Y high 8 bits | | | | |
| WzL | Angular velocity Z low 8 bits | Angular velocity Z=((WzH<<8)\|WzL)/32768*2000°/s | | | |
| WzH | Angular velocity Z high 8 bits | | | | |

| 0x55 | 0x53 | RollL | RollH | PitchL | PitchH |
|---|---|---|---|---|---|
| Name | Description | Remark | | | |
| RollL | Roll angle X lower 8 bits | Roll angle X= ((RollH<<8)\|RollL)/32768*180(°) | | | |
| RollH | Roll angle X high 8 bits | | | | |
| PitchL | Pitch angle Y low 8 bits | Pitch angle Y= ((PitchH<<8)\|PitchL)/32768*180(°) | | | |
| PitchH | Pitch angle Y high 8 bits | | | | |
| YawL | Yaw angle Z low 8 bits | Heading angle Z= ((YawH<<8)\|YawL)/32768*180(°) | | | |
| YawH | Yaw angle Z high 8 bits | | | | |

| 0x55 | 0x54 | HxL | HxH | HyL | HyH |
|---|---|---|---|---|---|
| Name | Description | Remark | | | |
| HxL | Magnetic field X lower 8 bits | Magnetic field X=((HxH<<8)\|HxL) | | | |
| HxH | Magnetic field X high 8 bits | | | | |
| HyL | Magnetic field Y lower 8 bits | Magnetic field Y=((HyH<<8)\|HyL) | | | |
| HyH | Magnetic field Y high 8 bits | | | | |
| HzL | Magnetic field Z lower 8 bits | Magnetic field Z=((HzH<<8)\|HzL) | | | |
| HzH | Magnetic field Z high 8 bits | | | | |

- Data is sent in hexadecimal, not ASCII.
- Each register address, the number of read registers, and the read data are represented by two bytes. The high and low bits of the register address are represented by ADDRH and ADDRL, the high and low bits of the number of registers to be read are represented by LENH and LENL, and the high and low bits of the read data are represented by DATA1H and DATA1L.

**command send**

| protocol header | protocol header | register | Data lower 8 bits | Data high 8 bits |
|---|---|---|---|---|
| 0xFF | 0xAA | ADDR | DATAL[7:0] | DATAH[15:8] |

**data return**

| 0x55 | 0x5F | REG1L | REG1H | REG2L | REG2H |
|---|---|---|---|---|---|
| Name | Description | Remark | | | |
| REG1L | Register 1 lower 8 bits | REG1[15:0] = ((REG1H<<8)\|REG1L) | | | |
| REG1H | Register 1 upper 8 bits | | | | |
| REG2L | Register 2 lower 8 bits | REG2[15:0] = ((REG2H<<8)\|REG2L) | | | |
| REG2H | Register 2 upper 8 bits | | | | |
| REG3L | Register 3 lower 8 bits | REG3[15:0] = ((REG3H<<8)\|REG3L) | | | |
| REG3H | Register 3 upper 8 bits | | | | |

- Data is sent in hexadecimal, not ASCII.
- All settings need to operate the unlock register (KEY) first.

- For each register address, the write data is represented by two bytes. The high and low bits of the register address are represented by ADDRH and ADDRL, and the high and low bits of the written data are represented by DATAH and DATAL.

| protocol header | protocol header | register | Data lower 8 bits | Data high 8 bits |
|---|---|---|---|---|
| 0xFF | 0xAA | ADDR | DATAL[7:0] | DATAH[15:8] |

# SAVE

| Register Name: SAVE Register Address: 0 (0x00) Read and write direction: R/W<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | SAVE[15:0] | Save: 0x0000<br>Reboot: 0x00FF Factory reset: 0x0001 |
| Example: FF AA 00 FF 00 (reboot) | | |

# CALSW

| | | |
|---|---|---|
| Register Name: CALSW Register Address: 1 (0x01) Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | CAL[3:0] | Set calibration mode:<br>0000(0x00): normal working mode 0001(0x01): Auto adder calibration 0011(0x03): height reset<br>0100(0x04): Set the heading angle to zero<br>0111(0x07): Magnetic Field Calibration (Spherical Fitting)<br>1000(0x08): set angle reference<br>1001(0x09): Magnetic Field Calibration (Dual Plane Mode) |
| Example: FF AA 01 04 00 (course angle set to zero) | | |

| | | |
|---|---|---|
| Register Name: RSW Register Address: 2 (0x02) Read and write direction: R/W<br>Default: 0x001E | | |
| Bit | NAME | FUNCTION |
| 15:11 | | |
| 10 | GSA (0x5A) | 0: off 1: on |
| 9 | QUATER (0x59) | 0: off 1: on |
| 8 | VELOCITY (0x58) | 0: off 1: on |
| 7 | GPS (0x57) | 0: off 1: on |
| 6 | PRESS (0x56) | 0: off 1: on |
| 5 | PORT (0x55) | 0: off 1: on |
| 4 | MAG (0x54) | 0: off 1: on |
| 3 | ANGLE (0x53) | 0: off 1: on |
| 2 | GYRO (0x52) | 0: off 1: on |
| 1 | ACC (0x51) | 0: off 1: on |
| 0 | TIME (0x50) | 0: off 1: on |
| Example: FF AA 02 3E 00 (set to output only acceleration, angular velocity, angle, magnetic field, port status) | | |

| Register Name: RRATE Register Address: 3 (0x03) Read and write direction: R/W<br>Default: 0x0006 | | |
| --- | --- | --- |
| Bit | NAME | FUNCTION |
| 15:4 | | |
| | | Set output rate |
| | | 0001(0x01): 0.2Hz |
| | | 0010(0x02): 0.5Hz |
| | | 0011(0x03): 1Hz |
| | | 0100(0x04): 2Hz |
| | | 0101(0x05): 5Hz |
| 3:0 | RRATE[3:0] | 0110(0x06): 10Hz |
| | | 0111(0x07): 20Hz |
| | | 1000(0x08): 50Hz |
| | | 1001(0x09): 100Hz |
| | | 1011(0x0B): 200Hz |
| | | 1011(0x0C): single return |
| | | 1100(0x0D): No return |
| Example：FF AA 03 03 00（set 1Hz output） | | |

**BAUD**

| | | |
|---|---|---|
| Register Name: BAUD Register Address: 4 (0x04) Read and write direction: R/W<br>Default: 0x0002 | | |
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | BAUD[3:0] | Set the serial port baud rate: 0000(0x00): 1000000bps<br>0001(0x01): 800000bps<br>0010(0x02): 500000bps<br>0011(0x03): 400000bps<br>0100(0x04): 250000bps<br>0101(0x05): 200000bps<br>0110(0x06): 125000bps<br>0111(0x07): 100000bps<br>1000(0x08): 80000bps<br>1001(0x09): 50000bps<br>1010(0x0A): 40000bps<br>1011(0x0B): 20000bps<br>1100(0x0C): 10000bps<br>1101(0x0D): 5000bps<br>1110(0x0E): 3000bps |
| Example: FF AA 04 02 00 (set serial port baud rate 500000) | | |

**AXOFFSET~HZOFFSET**

| Register Name: AXOFFSET~HZOFFSET Register Address: 5~13 (0x05~0x0D) Read and write direction: R/W<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | AXOFFSET[15:0] | Acceleration X-axis bias, actual acceleration offset=AXOFFSET[15:0]/10000(g) |
| 15:0 | AYOFFSET[15:0] | Acceleration Y-axis bias, actual acceleration offset=AYOFFSET[15:0]/10000(g) |
| 15:0 | AZOFFSET[15:0] | Acceleration Z-axis bias, actual acceleration offset=AZOFFSET[15:0]/10000(g) |
| 15:0 | GXOFFSET[15:0] | Angular velocity X-axis bias, actual angular velocity offset=GXOFFSET[15:0]/10000(°/s) |
| 15:0 | GYOFFSET[15:0] | Angular velocity Y-axis bias, actual angular velocity offset=GYOFFSET[15:0]/10000(°/s) |
| 15:0 | GZOFFSET[15:0] | Angular velocity Z-axis bias, actual angular velocity offset=GZOFFSET[15:0]/10000(°/s) |
| 15:0 | HXOFFSET[15:0] | Magnetic field X-axis zero bias |

| | | |
|---|---|---|
| 15:0 | HYOFFSET[15:0] | Magnetic field Y axis zero bias |
| 15:0 | HZOFFSET[15:0] | Magnetic field Z axis zero bias |
| Example: FF AA 05 E8 03 (set acceleration X-axis zero offset 0.1g), 0x03E8=1000, 1000/10000=0.1(g) | | |

## IICADDR

| | | |
|---|---|---|
| Register Name: IICADDR Register Address: 26 (0x1A) Read and write direction: R/W<br>Default: 0x0050 | | |
| Bit | NAME | FUNCTION |
| 15:8 | | |
| 7:0 | IICADDR[7:0] | Set the device address for I2C and Modbus communication<br>0x01~0x7F |
| Example: FF AA 1A 02 00 (set the device address to 0x02) | | |

| | | |
|---|---|---|
| Register Name: LEDOFF Register Address: 27 (0x1B) Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:1 | | |
| 0 | LEDOFF | 1: Turn off the LED light<br>0: Turn on the LED light |
| Example: FF AA 1B 01 00 (turn off the LED light) | | |

| | | |
|---|---|---|
| Register Name: MAGRANGX~MAGRANGZ Register Address: 28~30 (0x1C~0x1E) Read and write direction: R/W<br>Default: 0x01F4 | | |
| Bit | NAME | FUNCTION |
| 15:0 | MAGRANGX[15:0] | Magnetic field calibration X-axis range |
| 15:0 | MAGRANGY[15:0] | Magnetic Field Calibration Y-axis Range |
| 15:0 | MAGRANGZ[15:0] | Magnetic field calibration Z-axis range |
| Example: FF AA 1C F4 01 (set the magnetic field calibration X-axis range to 500) | | |

| Register Name: BANDWIDTH Register Address: 31 (0x1F) Read and write direction: R/W<br>Default: 0x0004 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | BANDWIDTH[3:0] | Set bandwidth 0000(0x00): 256Hz<br>0001(0x01): 188Hz<br>0010(0x02): 98Hz<br>0011(0x03): 42Hz<br>0100(0x04): 20Hz<br>0101(0x05): 10Hz<br>0110(0x06): 5Hz |
| Example：FF AA 1F 01 00 （set bandwidth is188Hz） | | |

| Register Name: GYRORANGE Register Address: 32 (0x20) Read and write direction: R/W<br>Default: 0x0003 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | GYRORANGE[3:0] | Set the gyro range 0011(0x03): 2000°/s<br>The default is 2000°/s, fixed and cannot be set |
| Example: FF AA 20 03 00 (set the gyro range to 2000°/s) | | |

| Register Name: ACCRANGE Register Address: 33 (0x21) Read and write direction: R/W<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | ACCRANGE[3:0] | Set the accelerometer range 0000(0x00): ±2g<br>0011(0x03): ±16g<br>This parameter cannot be set. The internal adaptive acceleration range of the product will automatically switch to 16g when the acceleration exceeds 2g. |
| Example: FF AA 21 00 00 (set the accelerometer range to 16g) | | |

| Register Name: SLEEP Register Address: 34 (0x22) Read and write direction: R/W<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:1 | | |
| 0 | SLEEP | set hibernate 1(0x01): sleep<br>Any serial port data, can wake up |
| Example: FF AA 22 01 00 (go to sleep) | | |

| | | |
|---|---|---|
| Register Name: ORIENT Register Address: 35 (0x23) Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:1 | | |
| 0 | ORIENT | Set the installation direction<br>0 (0x00): horizontal installation<br>1(0x01): vertical installation (the Y-axis arrow of the coordinate axis must be upward) |
| Example: FF AA 23 01 00 (set vertical installation) | | |

| | | |
|---|---|---|
| Register Name: AXIS6 Register Address: 36 (0x24) Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:1 | | |
| 0 | AXIS6 | set algorithm<br>0(0x00): 9-axis algorithm (magnetic field solution navigation angle, absolute heading angle)<br>1(0x01): 6-axis algorithm (integral solution navigation angle, relative heading angle) |
| Example: FF AA 24 01 00 (set 6-axis algorithm mode) | | |

| Register Name: FILTK Register Address: 37 (0x25) Read and write direction: R/W<br>Default: 0x001E | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | FILTK[15:0] | Range: 1~10000, the default is 30 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 30)<br>The smaller the FILTK[15:0], the stronger the seismic performance and the weaker the real-time performance.<br>The larger the FILTK[15:0], the weaker the seismic performance and the stronger the real-time performance. |
| Example: FF AA 25 1E 00 (set K value filter to 30) | | |

| | | |
|---|---|---|
| Register Name: READADDR Register Address: 39 (0x27) Read and write direction: R/W<br>Default: 0x00FF | | |
| Bit | NAME | FUNCTION |
| 15:8 | | |
| 7:0 | READADDR[7:0] | Read register range: Please refer to "Register Table" |
| Example:<br>Send: FF AA 27 34 00 (read acceleration X axis 0x34) Return: 55 5F AXL AXH AYL AYH AZL AZH GXL GXH SUM<br>For details, please refer to "Read Register Return Value" in the "Read Format" chapter | | |

| Register Name: ACCFILT Register Address: 42 (0x2A) Read and write direction: R/W<br><br>Default: 0x01F4 | | |
| --- | --- | --- |
| Bit | NAME | FUNCTION |
| 15:0 | ACCFILT[15:0] | Range: 1~10000, the default is 500 (it is not recommended to modify, once modified, if the angle does not meet the requirements for use, please modify it to 500)<br><br>The smaller the ACCFILT[15:0], the stronger the seismic performance and the weaker the real-time performance.<br><br>The larger the ACCFILT[15:0], the weaker the seismic performance and the stronger the real-time performance.<br><br>This parameter is an empirical value, which needs to be debugged according to different environments. In the tractor environment,<br><br>ACCFILT[15:0] can be adjusted to 100, because the vibration of the tractor is serious and the anti-vibration performance needs to be improved |

| | | |
|---|---|---|
| Example: FF AA 2A F4 01 (set acceleration filter 500) | | |

| | | |
|---|---|---|
| Register Name: POWONSEND Register Address: 45 (0x2D) Read and write direction: R/W<br>Default: 0x0001 | | |
| Bit | NAME | FUNCTION |
| 15:4 | | |
| 3:0 | POWONSEND[3:0] | Set the command to start:<br>0000(0x00): Turn off power-on data output<br>0001(0x01): Turn on power-on data output |
| Example: FF AA 2D 00 00 (turn on power-on data output) | | |

<table>
<tr><td>Register Name: VERSION Register Address: 46 (0x2E) Read and write direction: R<br><br>Default: none</td><td></td><td></td></tr>
<tr><td>Bit</td><td>NAME</td><td>FUNCTION</td></tr>
<tr><td>15:0</td><td>VERSION[15:0]</td><td>Different products, different version numbers</td></tr>
<tr><td>Example:<br><br>Send: FF AA 27 2E 00 (read version number, 0x27 means read, 0x2E is version number register)<br><br>Return: 55 5F VL VH XX XX XX XX XX XX SUM<br><br>VERSION[15:0]=(short)(((short)VH<<8)|VL)</td><td></td><td></td></tr>
</table>

| | | |
|---|---|---|
| Register Name: YYMM~MS<br>Register address: 48~51 (0x30~0x33) Read and write direction: R/W Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:8 | YYMM[15:8] | mouth |
| 7:0 | YYMM[7:0] | year |
| 15:8 | DDHH[15:8] | hour |
| 7:0 | DDHH[7:0] | day |
| 15:8 | MMSS[15:8] | seconds |
| 7:0 | MMSS[7:0] | minute |
| 15:0 | MS[15:0] | millisecond |
| Example:<br>FF AA 30 16 03 (set year 22-03)<br>FF AA 31 0C 09 (set date 12-09)<br>FF AA 32 1E 3A (set minute seconds 30:58) FF AA 33 F4 01 (set ms 500)<br>Example:<br>Send: FF AA 27 30 00 (read version number, 0x27 means read, 0x30 is year month | | |

register)

Returns: 55 5F YYMM[7:0] YYMM[15:8] DDHH[7:0] DDHH[15:8] MMSS[7:0] MMSS[15:8] MS[7:0] MS[15: 8] SUM

| Register Name: AX~AZ<br>Register address: 52~54 (0x34~0x36) Read and write direction: R<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | AX[15:0] | Acceleration X=AX[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2) |
| 15:0 | AY[15:0] | Acceleration Y=AY[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2) |
| 15:0 | AZ[15:0] | Acceleration Z=AZ[15:0]/32768*16g (g is the acceleration of gravity, preferably 9.8m/s2) |

| | | |
|---|---|---|
| Register Name: GX~GZ<br>Register address: 55~57 (0x37~0x39) Read and write direction: R<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | GX[15:0] | Angular velocity X=GX[15:0]/32768*2000 °/s |
| 15:0 | GY[15:0] | Angular velocity Y=GY[15:0]/32768*2000 °/s |
| 15:0 | GZ[15:0] | Angular velocity Z=GZ[15:0]/32768*2000 °/s |

| | | |
|---|---|---|
| Register name: HX~HZ<br>Register Address: 58~60 (0x3A~0x3C) Read and write direction: R<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | HX[15:0] | Magnetic field X=HX[15:0] (unit：LSB) |
| 15:0 | HY[15:0] | Magnetic field Y=HY[15:0] (unit：LSB) |
| 15:0 | HZ[15:0] | Magnetic fieldZ=HZ[15:0] (unit：LSB) |

| Register Name: Roll~Yaw<br>Register address: 61~63 (0x3D~0x3F) Read and write direction: R<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | Roll[15:0] | roll angleX=Roll[15:0]/32768*180° |
| 15:0 | Pitch[15:0] | picth angleY=Pitch[15:0]/32768*180° |
| 15:0 | Yaw[15:0] | heading angleZ=Yaw[15:0]/32768*180° |

| Register Name: TEMP Register Address: 64 (0x40) Read and write direction: R<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | TEMP[15:0] | temperature=TEMP[15:0]/100℃ |

| Register Name: PressureL~HeightH Register address: 69~72 (0x45~0x48) Read and write direction: R<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | PressureL[15:0] | air pressure= ((int)PressureH[15:0] <<16)\|PressureL[15:0] (Pa) |
| 15:0 | PressureH[15:0] | |
| 15:0 | HeightL[15:0] | altitute= ((int)HeightH[15:0] <<16)\|HeightL[15:0](cm) |
| 15:0 | HeightH[15:0] | |

| Register name: q0~q3<br>Register address: 81~84 (0x51~0x54) Read and write direction: R<br>Default: 0x0000 | | |
|---|---|---|
| Bit | NAME | FUNCTION |
| 15:0 | q0[15:0] | Quaternion0=q0[15:0]/32768 |
| 15:0 | q1[15:0] | Quaternion1=q1[15:0]/32768 |
| 15:0 | q2[15:0] | Quaternion=q2[15:0]/32768 |
| 15:0 | q3[15:0] | Quaternion3=q3[15:0]/32768 |

| | | |
|---|---|---|
| Register Name: GYROCALITHR Register Address: 97 (0x61) Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | GYROCALITHR[15:0] | Set the gyroscope inactivity threshold:<br>Gyro static threshold=GYROCALITHR [15:0]/1000(°/s) |
| Example: Setting the Gyro Rest Threshold to 0.05°/s FFAA 61 32 00<br>When the angular velocity change is less than 0.05°/s and lasts for the time of "GYROCALTIME", the sensor recognizes it as stationary and automatically resets the angular velocity less than 0.05°/s to zero<br>The setting rule of the static threshold of the gyroscope can be determined by reading the value of the "WERROR" register. The general setting rule is: GYROCALITHR=WERROR *1.2, unit: °/s<br>This register needs to be used in conjunction with the GYROCALTIME register | | |

| | | |
|---|---|---|
| Register Name: GYROCALTIME Register Address: 99 (0x63) Read and write direction: R/W<br>Default: 0x03E8 | | |
| Bit | NAME | FUNCTION |
| 15:0 | GYROCALTIME[15:0] | Set gyroscope auto-calibration time |
| Example: Set gyroscope auto-calibration time to 500ms FF AA 63 F4 01<br>When the angular velocity change is less than "GYROCALITHR" and lasts for 500ms, the sensor recognizes that it is stationary and automatically resets the angular velocity less than 0.05°/s to zero<br>This register needs to be used in conjunction with the GYROCALITHR register | | |

| | | |
|---|---|---|
| Register Name: KEY Register Address: 105 (0x69)<br>Read and write direction: R/W<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | KEY[15:0] | Unlock register: When performing a write operation, you need to set this register first |
| Example: Unlock, write 0xB588 to this register (other values are invalid)<br>FFAA 69 88 B5 | | |

| | | |
|---|---|---|
| Register Name: WERROR Register Address: 106 (0x6A) Read and write direction: R<br>Default: 0x0000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | WERROR[15:0] | Gyroscope change value=WERROR[15:0]/1000*180/3.1415926(°/s)<br>When the sensor is stationary, the "GYROCALITHR" register can be set by changing this register |

| | | |
|---|---|---|
| Register Name: WZTIME Register Address: 110 (0x6E) Read and write direction: R/W<br>Default: 0x01F4 | | |
| Bit | NAME | FUNCTION |
| 15:0 | WZTIME[15:0] | Angular velocity continuous rest time |
| Example: Set the angular velocity continuous static time to 500ms FFAA 6E F4 01<br>When the angular velocity is less than "WZSTATIC" for 500ms, the angular velocity output is 0, and the Z-axis heading angle is not integrated<br>This register needs to be used in conjunction with the "WZSTATIC" register | | |

| | | |
|---|---|---|
| Register Name: WZSTATIC Register Address: 111 (0x6F) Read and write direction: R/W<br>Default: 0x012C | | |
| Bit | NAME | FUNCTION |
| 15:0 | WZSTATIC[15:0] | Angular velocity integration threshold=WZSTATIC[15:0]/1000(°/s) |
| Example: Set the angular velocity integration threshold to 0.5°/s FF AA 6F F4 01<br>When the angular velocity is greater than 0.5°/s, the Z-axis heading angle starts to integrate the acceleration<br>When the angular velocity is less than 0.5°/s, and the duration set by the register "WZTIME", the angular velocity output is 0, and the Z-axis heading angle is not integrated<br>This register needs to be used in conjunction with the "WZTIME" register | | |

| | | |
|---|---|---|
| Register Name: XREFROLL~YREFPITCH<br>Register address: 121~122 (0x79~0x7A)<br>Read and write direction: R/W<br>Default: 0x00000 | | |
| Bit | NAME | FUNCTION |
| 15:0 | XREFROLL[15:0] | Roll angle zero reference value=XREFROLL[15:0]/32768*180(°) |
| 15:0 | YREFPITCH[15:0] | Pitch angle zero reference value=YREFPITCH[15:0]/32768*180(°) |
| Example: The current roll angle is 2°, set the roll angle zero, subtract 2°, then XREFROLL[15:0]=2*32768/180=364=0x016C<br>FFAA 79 6C 01 | | |

| | | |
|---|---|---|
| Register Name: NUMBERID1~NUMBERID6 Register address: 127~132 (0x7F~0x84) Read and write direction: R<br>Default: none | | |
| Bit | NAME | FUNCTION |
| 15:0 | NUMBERID1[15:0] | |
| 15:0 | NUMBERID2[15:0] | |
| 15:0 | NUMBERID3[15:0] | |
| 15:0 | NUMBERID4[15:0] | |
| 15:0 | NUMBERID5[15:0] | |
| 15:0 | NUMBERID6[15:0] | |
| Device label：WT4200000001 | | |

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